

Counting positive rational friezes over $\frac{1}{N}\mathbb{Z}$

Vico Bonfioli

vicobonfioli@gmail.com

Abstract

Let $T(N, n)$ denote the number of positive rational frieze patterns of width n all of whose entries lie in $\frac{1}{N}\mathbb{Z}$. Conway and Coxeter showed $T(1, n) = C_{n-2}$. We prove that for every prime $p \geq 5$

$$T(p, 5) = 5 + 5C(p), \quad C(p) = \#\{(a, u, v) \in \mathbb{Z}_{>0}^3 : auv = p + u + v\},$$

an equality identifying the width-five count with the case $A = 1$, $B = p$ of $xyz = A(x+y) + B$, a problem posed by Ford in generalisation of a question of Conrey. The order of $C(p)$ is open, so no closed form for $T(p, 5)$ should be expected.

At every N the count decomposes the same way. It is a finite sum of counts for Ford's equation, one term for each divisor M of N^2 with $(N/\gcd(N, M)) \mid M$, the term carrying the parameters $A = \gcd(N, M)$ and $B = M \gcd(N, M)/N$, and the number of terms is the number of square divisors of N^3 . At a prime the sum has two terms and collapses to the display above; for composite N it does not collapse, and no single instance of Ford's equation carries the count.

We also prove $T(N, 4) = d(2N^2)$, give proved search bounds at widths 5 and 6, so that the tabulated values are complete rather than observed to stabilise, and tabulate $T(N, n)$ for widths 3 to 6 and, at small N , for widths 8 and 9. The rotation action of $\mathbb{Z}/5$ on width-5 quiddity cycles is free, so $5 \mid T(N, 5)$. We prove $T(N, 5) = O_\varepsilon(N^{1+\varepsilon})$, and $T(p, 5) = O_\varepsilon(p^{1/3+\varepsilon})$ at primes by importing the method of Conrey and Shah, while $T(p, 5) \geq 5d(p+1)$ refutes the guess $T(N, 5) \asymp d(N^2) \log^2 N$.

The reduction behind the tables holds at every width: homogenising the frieze continuant and applying the Desnanot–Jacobi identity collapses the lattice conditions to $gUV = N^{n-4}(U+V) + M$ with $M \mid N^2R$ and $N^{n-4}R \mid gM$, where R is a continuant window empty at width 5.

The count at a prime is machine-checked in Lean 4 from the definition of a frieze itself, with periodicity derived rather than assumed; Section 10 says what else is verified and what is not.

1 Introduction

A frieze pattern of width n is a finite array of rationals bordered by zeros and ones in which every adjacent 2×2 minor equals 1. Conway and Coxeter [4] classified the patterns with positive integer entries: they correspond to triangulations of a convex n -gon, so there are C_{n-2} of them. Allowing denominators changes the problem. Fixing N and asking for the positive patterns with all entries in $\frac{1}{N}\mathbb{Z}$ gives a count $T(N, n)$, finite because $\frac{1}{N}\mathbb{Z}$ is discrete and Cuntz and Holm [3] showed a discrete subset admits only finitely many friezes of each height, and computing it is the subject of this paper.

The point of this paper is that this count is an open problem in analytic number theory wearing different clothes. At a prime,

$$T(p, 5) = 5 + 5C(p), \quad C(p) = \#\{(a, u, v) \in \mathbb{Z}_{>0}^3 : auv = p + u + v\}$$

(Theorem 7.7). The width-5 reduction is $guv = N(u + v) + M$, which is $xyz = A(x + y) + B$ with $A = N$ and $B = M$, posed by Ford in generalisation of a question of Conrey and recorded in [9]; the cubic $auv = p + u + v$ above is its case $A = 1$, $B = p$. Ford appears to have published nothing further on it. The fully symmetric variant $n = xyz + x + y + z$, also asked by Conrey, is Problem 25 of the American Institute of Mathematics problem list of May 2026 and is the equation of Conrey and Shah [7]. The order of $C(p)$ is not known. Counting friezes over $\frac{1}{p}\mathbb{Z}$ and estimating $C(p)$ are the same problem.

Friezes have been related to equations of this shape before, and the difference matters. Zhang [1] relates friezes to $xyz = G(x, y)$ in a different setting and direction: positive *integral* friezes of Dynkin type, where the Fontaine–Plamondon counts are determined exactly, and where the Diophantine conclusion is the existence of infinitely many positive solutions rather than a bound on their number. The equations here are the rational ones over $\frac{1}{N}\mathbb{Z}$, and the question is the counting bound asked by Conrey and, in generalised form, by Ford. Conley and Ovsienko [2] enumerate the positive *integer* solutions of the same matrix equation, graded by a parameter measuring the departure of a dissection from a triangulation; that parameter and the denominator N are independent, and none of the sequences below occurs in their tables.

Theorem 7.7 is an equality, not an estimate, so an asymptotic for either side transfers to the other. Neither side is settled here. The count is $O_\varepsilon(p^{1/3+\varepsilon})$ by the method of Conrey and Shah [7] (Theorem 9.3), against a conjectured $p^{o(1)}$ (Conjecture 9.4). The lower bounds of Propositions 9.5 and 9.6, $T(p, 5) \geq 5d(p + 1)$ and $C(p) \geq d(p + 1) + d(2p + 1)$, are the $a = 1$ and $a = 2$ slices; Conrey and Shah record the same two for their equation in [7, §2.1]. Proposition 9.7 puts the count on the modular hyperbola $de \equiv 1 \pmod{p}$, where the Weil error is of size $p^{1/2}$ against a quantity of size $p^{1/3}$, which is why the standard route does not reach even the trivial bound.

For frieze theory this explains an absence. $T(N, 4) = d(2N^2)$ has a closed form, counts of friezes over $\mathbb{Z}/n\mathbb{Z}$ are multiplicative with explicit formulas, and no such formula has been found at width 5 over $\frac{1}{N}\mathbb{Z}$: finding one would settle a question of Conrey. Karpenkov, Short, van Son and Zabolotskii [5] classified exactly these friezes and left the count open. In the other direction it gives $C(p)$ a combinatorial reading, as the count of a set of friezes, which is not how the quantity in Ford’s problem arises elsewhere.

Section 7 tabulates $T(N, n)$ for widths 3 to 6 and for widths 8 and 9 at small N . At width 4 the count has a closed form, $T(N, 4) = d(2N^2)$ (Theorem 4.1). The reductions and search bounds behind the tables at the larger widths are routine and are carried out in the accompanying library rather than here.

Section 10 says which results are machine-checked and which rest on a written proof; the per-statement map to Lean declarations is maintained with the library. Tables and computed constants are regenerated by scripts in `code/`, named at the point of use; `code/paper_numbers.py` collects those that have no script of their own.

2 Setting

We use the conventions of Karpenkov, Short, van Son and Zabolotskii [5].

Definition 2.1. A *rational frieze of width n* is a map $F: \{(i, j) : 0 \leq i - j \leq n\} \rightarrow \mathbb{Q}$, written $m_{i,j}$, with $m_{i,j} = 0$ when $i - j \in \{0, n\}$, with $m_{i,j} = 1$ when $i - j \in \{1, n - 1\}$, and satisfying the diamond rule

$$m_{i,j} m_{i+1,j+1} - m_{i,j+1} m_{i+1,j} = 1.$$

It is *positive* if every entry with $0 < i - j < n$ is positive. The *quiddity cycle* is the row $i - j = 2$, a sequence $(a_0, \dots, a_{n-1})$ read cyclically.

The quiddity cycle determines the frieze: writing r for the row index,

$$m_{r,j} = \frac{m_{r-1,j} m_{r-1,j+1} - 1}{m_{r-2,j+1}}, \quad (1)$$

with the closing conditions that row $n - 1$ is constant 1 and row n is constant 0.

Definition 2.2. For $N \geq 1$ let $T(N, n)$ be the number of positive rational friezes of width n all of whose entries lie in $\frac{1}{N}\mathbb{Z}$. Quiddity cycles are counted as sequences, not up to rotation.

Conway and Coxeter [4] showed that the positive *integer* friezes of width n correspond to triangulations of a convex n -gon, so

$$T(1, n) = C_{n-2} = \frac{1}{n-1} \binom{2n-4}{n-2}. \quad (2)$$

Finiteness of $T(N, n)$ is Corollary 3.8 of Cuntz and Holm [3], which gives it for any discrete $R \subseteq \mathbb{C}$. Theorem B of [5] classifies the positive rational friezes over $\frac{1}{N}\mathbb{Z}$ in terms of minimal clockwise paths in a Farey-type graph. It classifies exactly the objects counted here, and does not count them. The present note is concerned with computing it.

3 The continuant parameterisation

Let K_k denote the frieze continuant, $K_k = x_k K_{k-1} - K_{k-2}$ with $K_0 = 1$, $K_{-1} = 0$. Continuants and the continued-fraction geometry behind them are treated in [6].

Proposition 3.1. A positive rational frieze of width $n \geq 4$ is determined by $a_0, \dots, a_{n-4}$, and

$$\begin{aligned} a_{n-3} &= \frac{K_{n-4}(a_0, \dots, a_{n-5}) + 1}{K_{n-3}(a_0, \dots, a_{n-4})}, & a_{n-2} &= K_{n-3}(a_0, \dots, a_{n-4}), \\ a_{n-1} &= \frac{K_{n-4}(a_1, \dots, a_{n-4}) + 1}{K_{n-3}(a_0, \dots, a_{n-4})}. \end{aligned}$$

Proof. The closing conditions are $m_{n-1,j} = 1$ and $m_{n,j} = 0$ for every j , that is $K_{n-2}(a_j, \dots, a_{j+n-3}) = 1$ and $K_{n-1}(a_j, \dots, a_{j+n-2}) = 0$. Expanding the second at $j = 0$ by the recursion $K_k = x_k K_{k-1} - K_{k-2}$ in its last argument and solving for a_{n-2} gives the middle display; expanding the first at $j = 0$ and at $j = 1$ and solving for a_{n-3} and a_{n-1} gives the other two. The remaining $n - 3$ closing conditions are cyclic conjugates of these under the monodromy $M(a_0) \cdots M(a_{n-1}) = -I$ and impose nothing further. $\square$

At $n = 5$ this reads

$$a_2 = \frac{a_0 + 1}{D}, \quad a_3 = D, \quad a_4 = \frac{a_1 + 1}{D}, \quad D = a_0 a_1 - 1, \quad (3)$$

and at $n = 6$, with $D = K_3(a_0, a_1, a_2) = a_2(a_0a_1 - 1) - a_0$,

$$a_3 = \frac{a_0a_1}{D}, \quad a_4 = D, \quad a_5 = \frac{a_1a_2}{D}. \quad (4)$$

Remark 3.2. A frieze of width n has a glide symmetry exchanging row r with row $n - r$. For $n = 5$ the only non-constant interior rows are 2 and 3, and they are exchanged, so a quiddity cycle in $\frac{1}{N}\mathbb{Z}$ forces every entry into $\frac{1}{N}\mathbb{Z}$. For $n = 6$ row 3 is fixed by the glide, and carries a lattice condition of its own. This is the source of the extra divisibility constraints at width 6.

4 Width four

Write d for the number-of-divisors function.

Theorem 4.1. *For every $N \geq 1$, $T(N, 4) = d(2N^2)$.*

Proof. Let (a_0, a_1, a_2, a_3) be the quiddity cycle. By (1) the row 3 entries are $a_j a_{j+1} - 1$, and row 3 is constant 1, so $a_j a_{j+1} = 2$ for all j modulo 4. Hence $a_0 = a_2$, $a_1 = a_3$ and $a_0 a_1 = 2$. Conversely any $a > 0$ with $a \cdot (2/a) = 2$ yields the frieze with quiddity $(a, 2/a, a, 2/a)$, whose remaining rows are forced and positive.

Write $a_0 = p/N$ with p a positive integer. Then $a_1 = 2N/p$, and $a_1 \in \frac{1}{N}\mathbb{Z}$ if and only if $2N^2/p \in \mathbb{Z}$, that is $p \mid 2N^2$. Distinct divisors give distinct cycles, so $T(N, 4)$ is the number of positive divisors of $2N^2$. $\square$

The sequence $d(2N^2)$ is OEIS A361689.

5 Width five

Put $a_0 = p/N$ and $a_1 = q/N$ with p, q positive integers, and set $e := pq - N^2$.

Proposition 5.1. *The parameters (p, q) arise from a positive frieze of width 5 over $\frac{1}{N}\mathbb{Z}$ if and only if*

$$e > 0, \quad N \mid e, \quad e \mid N^2(p + N), \quad e \mid N^2(q + N). \quad (5)$$

Proof. By (3), $a_3 = D = e/N^2$, $a_2 = N(p + N)/e$ and $a_4 = N(q + N)/e$. Positivity of the frieze is equivalent to $D > 0$, that is $e > 0$. Membership of a_3 in $\frac{1}{N}\mathbb{Z}$ is $N \mid e$, and membership of a_2 and a_4 is the two divisibility conditions. By Remark 3.2 no further condition is needed. $\square$

Lemma 5.2 (Search bound). *Any solution of (5) satisfies $p \leq N^3 + 2N^2$ and $q \leq N^3 + 2N^2$.*

Proof. Since $e > 0$ and $e \mid N^2(q + N)$ with $N^2(q + N) > 0$, we have $e \leq N^2(q + N)$. Substituting $e = pq - N^2$ gives $pq - N^2 \leq N^2q + N^3$, that is $q(p - N^2) \leq N^3 + N^2$. As $q \geq 1$, either $p \leq N^2$ or $p - N^2 \leq N^3 + N^2$; in both cases $p \leq N^3 + 2N^2$. The bound on q is symmetric. $\square$

Lemma 5.2 makes the enumeration finite with an explicit range. Since additionally $e \mid N^2(p + N)$ and $q = (e + N^2)/p$, one may loop over p in the proved range and over the divisors e of $N^2(p + N)$, which is what `code/verify_width5_proved.py` does. The resulting values of $T(N, 5)$ for $N \leq 40$ are recorded in `data/T5.txt`. They are complete, not merely stable under an increasing cutoff.

6 Symmetry at width five

The dihedral group D_5 acts on width-5 quiddity cycles by rotation and reversal.

Theorem 6.1. *The rotation action of $\mathbb{Z}/5$ on positive rational width-5 quiddity cycles is free. In particular $5 \mid T(N, 5)$ for every N .*

Proof. The closing condition at width 5 is $(a_j a_{j+1} - 1)(a_{j+1} a_{j+2} - 1) = 1 + a_{j+1}$ for all j modulo 5. A cycle fixed by a nontrivial rotation is constant, say $a_j = a$, whence $(a^2 - 1)^2 = 1 + a$, that is $a(a^3 - 2a - 1) = 0$. Since $a > 0$ and $a^3 - 2a - 1 = (a + 1)(a^2 - a - 1)$, the only positive root is the golden ratio $(1 + \sqrt{5})/2$, which is irrational. So no positive rational cycle is fixed. $\square$

7 Data and questions

Every value below is produced by a procedure whose search range is proved, so the counts are complete rather than stable under an increasing cutoff.

	$n = 3$	$n = 4$	$n = 5$	$n = 6$
$N = 1$	1	2	5	14
$N = 2$	1	4	20	102
$N = 3$	1	6	40	259
$N = 4$	1	6	60	487

Neither $T(N, 5)$ nor $T(N, 6)$ appears in the OEIS as a function of N , and neither is multiplicative: $T(2, 5)T(3, 5) = 800 \neq 110 = T(6, 5)$.

At widths 8 and 9 the same reduction, applied with the glide symmetry as a pruning rule and swept over the entry box of Cuntz and Holm [3], gives

N	1	2	3	4
$T(N, 8)$	132	2576	9980	28604
$T(N, 9)$	429	13101		

(code/w89_count.rs). The values at $N = 1$ are C_6 and C_7 , as they must be, and neither sequence appears in the OEIS. The sweep is complete over a proved box, so these are counts rather than stable observations; the cost grows like $N^{3(n-3)}$, which is why the table stops where it does. The reduction and the search bound behind it are routine and are carried out in the accompanying library rather than here.

Theorem 7.1 (Reduced form). *For every $N \geq 1$,*

$$T(N, 5) = \#\{(u, v, M) \in \mathbb{Z}_{>0}^3 : \gcd(u, v) = 1, M \mid N^2, \\ uv \mid N(u + v) + M, Nuv \mid M(N(u + v) + M)\}.$$

Proof. Multiplying $gu > N$ by $v > 0$ gives $guv > Nv$, that is $Nu + M > 0$, which holds for $u, M > 0$; the same argument disposes of $gv > N$. So both positivity conditions are automatic. For the equation, g is determined by (u, v, M) , so requiring some g with $guv = N(u + v) + M$ is requiring $uv \mid N(u + v) + M$. Given that, multiplying $N \mid gM$ by uv gives $Nuv \mid M(N(u + v) + M)$, and the two are equivalent because $uv \neq 0$. $\square$

Proposition 7.2 (Hyperbola bound). *Every solution counted by Theorem 7.1 satisfies $(u - N)(v - N) \leq N^2 + M \leq 2N^2$, and if $u \leq v$ then $u < 3N$.*

Proof. Since uv divides the positive integer $N(u + v) + M$ it is at most that integer, and $(u - N)(v - N) = uv - N(u + v) + N^2$, so $(u - N)(v - N) \leq N^2 + M$; and $M \leq N^2$ because $M \mid N^2$. If $u \leq N$ then $u < 3N$. Otherwise $0 < u - N \leq v - N$, so $(u - N)^2 \leq (u - N)(v - N) \leq 2N^2$, whereas $u \geq 3N$ would force $(u - N)^2 \geq 4N^2$. $\square$

The change is one of order: the smaller parameter is bounded *linearly* in N , where Lemma 5.2 gives only $N^3 + 2N^2$ for the original parameters. The bound $(u - N)(v - N) \leq 2N^2$ is attained for every $2 \leq N \leq 24$ (`code/t5_bounds.py`).

Corollary 7.3. $T(N, 5) \leq 2 \sum_{M \mid N^2} \sum_{1 \leq u < 3N} d(Nu + M)$, and consequently $T(N, 5) = O_\varepsilon(N^{1+\varepsilon})$.

Proof. By Proposition 7.2 the smaller of u, v is less than $3N$; fix it as u and let v run over the divisors of $Nu + M$, which is legitimate because $v \mid N(u + v) + M$ and $v \mid Nv$ together give $v \mid Nu + M$. Each unordered pair is counted at most twice. For the second statement, $d(n) = O_\varepsilon(n^\varepsilon)$ gives $d(N^2) = O_\varepsilon(N^\varepsilon)$ and $d(Nu + M) = O_\varepsilon(N^\varepsilon)$ uniformly for $u < 3N$ and $M \leq N^2$, so the double sum is $O_\varepsilon(N^{1+\varepsilon})$. $\square$

No matching lower bound of the same order appears to exist.

Definition 7.4. Let p be prime. In the parametrisation of Theorem 7.1 the parameter M divides p^2 , so $M \in \{1, p, p^2\}$, and $M = 1$ cannot occur: the condition $Nuv \mid M(N(u + v) + M)$ reads $puv \mid p(u + v) + 1$ there, forcing $p \mid 1$. Write $A(p)$ for the number of solutions with $M = p$ and $B(p)$ for the number with $M = p^2$, so that $T(p, 5) = A(p) + B(p)$. At $M = p$ both divisibility conditions of Theorem 7.1 reduce to $uv \mid p(u + v + 1)$, so

$$A(p) = \#\{(u, v) \in \mathbb{Z}_{>0}^2 : \gcd(u, v) = 1, uv \mid p(u + v + 1)\}.$$

A quiddity cycle has $M = p$ exactly when its parameters do, so this is also the count of cycles with $M = p$ used in Theorem 7.7.

Proposition 7.5. *Let $p, u, v \geq 1$ with $\gcd(uv, p) = 1$ and $uv \mid p(u + v + 1)$. Then*

$$(u, v) \in \{(1, 1), (1, 2), (2, 1), (2, 3), (3, 2)\},$$

independently of p . These five pairs form a single orbit of the rotation action of $\mathbb{Z}/5$ of Theorem 6.1; that last assertion is not proved here and nothing below depends on it.

Proof. Since $u \mid uv \mid p(u + v + 1)$ and $\gcd(u, p) = 1$ we get $u \mid u + v + 1$, hence $u \mid v + 1$; symmetrically $v \mid u + 1$. So $u \leq v + 1$ and $v \leq u + 1$. If $u = v$ then $u \mid u + 1$ gives $u = 1$. If $v = u + 1$ then $u \mid v + 1 = u + 2$ gives $u \mid 2$, so (u, v) is $(1, 2)$ or $(2, 3)$; the case $u = v + 1$ is symmetric. Coprimality of u and v is not needed. $\square$

So every further solution at a prime requires $p \mid uv$, and $T(p, 5)$ stays small: it is 160 at $p = 541$ and 130 at $p = 421$, against $T(120, 5) = 4170$.

By Remark 3.2 the width-5 glide gives $a_{j+3} = a_j a_{j+1} - 1$ for every j modulo 5, which in numerators is

$$p_j p_{j+1} = p(p_{j+3} + p) \quad (j \in \mathbb{Z}/5), \quad (6)$$

so $e_j = pp_{j+3}$ and therefore $M_j = pp_{j+3}/g_j$ with $g_j = \gcd(p_j + p, p_{j+1} + p)$. In particular $M_j = p^2$ forces $p \mid p_{j+3}$, so the M -profile is controlled by which entries of the quiddity are integers.

Proposition 7.6. *Let p be prime. The number of integer entries in the quiddity cycle of a positive width-5 frieze over $\frac{1}{p}\mathbb{Z}$ is either 3 or 5, and it is 5 exactly for the Conway–Coxeter frieze $(2, 2, 1, 3, 1)$ and its rotations.*

Proof. Put $I = \{j : p \mid p_j\}$. If $p \nmid p_{j+3}$ then the right side of (6) has p -valuation exactly one, so exactly one of p_j, p_{j+1} is divisible by p , and that one has valuation exactly one. Taking $j = m + 2$, this says: for every $m \notin I$, exactly one of $m + 2, m + 3$ lies in I . Exhausting the 32 subsets of $\mathbb{Z}/5$, the only ones with that property have cardinality 3 or 5, and those of cardinality 3 are exactly the sets $\{i, i + 1, i + 3\}$. If $|I| = 5$ every entry is an integer, so the frieze is a Conway–Coxeter frieze of width 5, and there are $C_3 = 5$ of those, forming one rotation orbit. $\square$

Theorem 7.7 (Orbit split). *Let p be prime and write $R = T(p, 5)/5$ for the number of rotation orbits. Each orbit contains exactly two cycles with $M = p$ and three with $M = p^2$, except the orbit of the Conway–Coxeter frieze $(2, 2, 1, 3, 1)$, where all five have $M = p$. Hence $A(p) = 2R + 3$ and $B(p) = 3R - 3$, so that*

$$3A(p) - 2B(p) = 15 \quad \text{and} \quad T(p, 5) = \frac{1}{2}(5A(p) - 15).$$

Proof. By Proposition 7.6, $I = \{j : p \mid p_j\}$ has three or five elements. If it has five the frieze is $(2, 2, 1, 3, 1)$ up to rotation, and evaluating M_j at each of the five positions gives p every time.

Suppose $|I| = 3$. The condition established in the proof of Proposition 7.6, that every $m \notin I$ has exactly one of $m + 2, m + 3$ in I , forces $I = \{i, i + 1, i + 3\}$ for some i . That proof also records that the divisible one of p_j, p_{j+1} has valuation exactly one whenever $p \nmid p_{j+3}$; applying this at $j = i - 1$ and at $j = i + 1$, where p_{i+2} respectively p_{i+4} is prime to p , gives $v_p(p_i) = v_p(p_{i+1}) = 1$ exactly, while $v_p(p_{i+2}) = v_p(p_{i+4}) = 0$ because $i + 2, i + 4 \notin I$. Now take the five positions in turn.

For $j = i + 1$ and $j = i + 4$ we have $j + 3 \notin I$. In each case one of the two entries p_j, p_{j+1} is prime to p , hence so is g_j , and $v_p(M_j) = 1 + 0 - 0 = 1$, so $M_j = p$.

For $j = i + 2$ and $j = i + 3$ the entry p_{i+2} respectively p_{i+4} is prime to p , so again g_j is prime to p , while $v_p(p_{j+3}) = 1$; thus $v_p(M_j) = 2$ and $M_j = p^2$.

For $j = i$ write $p_i = p\alpha$ and $p_{i+1} = p\beta$, so $g_i = pG$ with $G = \gcd(\alpha + 1, \beta + 1)$, and (6) at $j = i$ gives $p_{i+3} = p(\alpha\beta - 1)$ with $\alpha\beta - 1 > 0$. If $M_i = p$ then $g_i = p_{i+3}$, that is $G = \alpha\beta - 1$. Since $G \mid \alpha + 1$ this makes $\alpha\beta - 1$ divide $\alpha + 1$, say $\alpha + 1 = (\alpha\beta - 1)s$. But (6) at $j = i + 2$ reads $p_{i+2}p_{i+3} = p(p_i + p)$, that is $p_{i+2}(\alpha\beta - 1) = p(\alpha + 1)$, so $p_{i+2}(\alpha\beta - 1) = p(\alpha\beta - 1)s$, and cancelling $\alpha\beta - 1 \neq 0$ gives $p_{i+2} = ps$, contradicting $p \nmid p_{i+2}$. Hence $M_i = p^2$.

So two positions carry $M = p$ and three carry $M = p^2$. Counting over the R orbits, $A(p) = 2(R - 1) + 5$ and $B(p) = 3(R - 1)$, and $T(p, 5) = A + B = 5R$. $\square$

8 The count at every N

Theorem 7.1 counts triples (u, v, M) with M running over the divisors of N^2 . The parameter can be removed from each term at the cost of a sum: what remains at each contributing M is an instance of the equation of Ford’s problem.

Definition 8.1. For $A, B \geq 1$ put

$$F(A, B) = \#\{(u, v) \in \mathbb{Z}_{>0}^2 : \gcd(u, v) = 1, uv \mid A(u + v) + B\},$$

the number of coprime solutions of $huv = A(u + v) + B$, the equation of Ford's problem with those parameters.

Theorem 8.2. Let $N \geq 1$. For $M \mid N^2$ write $e = \gcd(N, M)$ and $d = N/e$. Then

$$T(N, 5) = \sum_{\substack{M \mid N^2 \\ d \mid M}} F\left(e, \frac{M}{d}\right),$$

and the divisors omitted from the sum contribute nothing: no solution counted by Theorem 7.1 has $d \nmid M$.

Proof. Fix $M \mid N^2$ and a coprime pair (u, v) . The first condition of Theorem 7.1 says that $g := (N(u + v) + M)/uv$ is an integer, and given that, the second is $N \mid gM$. Write $M = eM'$; then $\gcd(d, M') = 1$, so $N \mid gM$ is $ed \mid geM'$, that is $d \mid gM'$, that is $d \mid g$.

Put $g = dh$. Then $dhu v = ed(u + v) + M$, and since $d \mid ed(u + v)$ this forces $d \mid M$, which is the support condition; dividing by d leaves

$$huv = e(u + v) + \frac{M}{d}. \quad (7)$$

Conversely, if $\gcd(u, v) = 1$ and (7) holds for some $h \geq 1$, then $g := dh$ satisfies $guv = N(u + v) + M$ and $d \mid g$, hence $N \mid gM$. The positivity conditions are automatic, by the argument of Theorem 7.1. So for each M in the support the pairs counted by Theorem 7.1 correspond bijectively to those counted by $F(e, M/d)$, and the sum over M gives the identity. $\square$

Corollary 8.3. Write $N = \prod_p p^{a_p}$. The sum of Theorem 8.2 has $\prod_p ([3a_p/2] + 1)$ terms, the number of square divisors of N^3 (OEIS A092520).

Proof. For $N = p^a$ and $M = p^m$ one has $d = p^{\max(a-m, 0)}$, so $d \mid M$ reads $[a/2] \leq m$, and $M \mid N^2$ reads $m \leq 2a$. That leaves $2a - [a/2] + 1 = [3a/2] + 1$ values of m . Both the divisor set and the condition are multiplicative in N . $\square$

At a prime the sum has two terms. The divisors of p^2 are $1, p, p^2$; for $M = 1$ one has $d = p$, which does not divide 1, while $M = p$ and $M = p^2$ both give $e = p$ and $d = 1$. So

$$T(p, 5) = F(p, p) + F(p, p^2) = A(p) + B(p),$$

the split of Definition 7.4, and Theorem 7.7 evaluates it as $5 + 5C(p)$. That collapse is special to primes. For composite N the sum has more terms and no single instance of Ford's equation carries the count; in particular the guess $T(N, 5) = 5 + 5C(N)$, with $C(N)$ the solution count of $auv = N + u + v$, is false, already at $N = 12$, where the two sides are 310 and 40.

9 The width-five count at a prime

No closed form for $T(p, 5)$ should be expected. Setting $u = ps$ in the $M = p$ term and writing $v + 1 = st$, the condition $uv \mid p(u + v + 1)$ becomes $st - 1 \mid p + t$, and putting $p + t = (st - 1)m$ turns it into the symmetric cubic

$$stm - t - m = p, \quad \text{equivalently} \quad (st - 1)(sm - 1) = sp + 1. \quad (8)$$

The correspondence is exact.

Lemma 9.1. *Let $p \geq 5$ be prime and let $C(p)$ be the number of triples (s, t, m) of positive integers with $stm = p + t + m$. Then $A(p) = 5 + 2C(p)$.*

Proof. By Definition 7.4, $A(p)$ counts the coprime pairs (u, v) with $uv \mid p(u + v + 1)$, a condition symmetric in u and v . If $\gcd(uv, p) = 1$ it reduces to $uv \mid u + v + 1$, and Proposition 7.5 lists the five pairs satisfying it. Otherwise $p \mid uv$, and $\gcd(u, v) = 1$ forces p to divide exactly one of u and v ; by the symmetry the two halves are equinumerous, so $A(p) - 5 = 2 \#\{(u, v) : p \mid u\}$.

For the pairs with $p \mid u$ write $u = ps$. Then $uv \mid p(u + v + 1)$ reads $sv \mid ps + v + 1$, whence $s \mid v + 1$; put $v = st - 1$ with $t \geq 1$. The condition becomes $s(st - 1) \mid s(p + t)$, that is $st - 1 \mid p + t$; put $p + t = (st - 1)m$ with $m \geq 1$. Expanding, $stm = p + t + m$, which is (8).

Conversely, given (s, t, m) with $stm = p + t + m$, set $u = ps$ and $v = st - 1$. Then $st - 1 = (p + t)/m \geq 1$, so $v \geq 1$, and $\gcd(u, v) = 1$: indeed $\gcd(s, st - 1) = 1$, and $p \nmid st - 1$, since $st \equiv 1 \pmod{p}$ would give $m \equiv t + m$, hence $p \mid t$ and then $st \equiv 0$, a contradiction. The two maps are mutually inverse, so $\#\{(u, v) : p \mid u\} = C(p)$. $\square$

Checked against $A(p)$ computed directly for the primes $5 \leq p \leq 53$ (`code/a_count.py`), with no discrepancy. Equation (8) also bounds the count, and does so better than Corollary 7.3.

The exponent can be lowered to $\frac{1}{3}$ by importing the method of Conrey and Shah [7], who bound the number of representations $n = xyz + x + y + z$ by $n^{1/3} \log n (\log \log n)^4$. Their argument uses that full symmetry forces the smallest variable below $n^{1/3}$. Our equation $suv = p + u + v$ is symmetric only in u and v , but the same bound holds.

Lemma 9.2. *For $p \geq 4$, every solution of $suv = p + u + v$ in positive integers has $\min(s, u, v)^3 \leq 2p$.*

Proof. If $\min(s, u, v)^3 > 2p$ then $suv > 2p$, so $p < u + v$ and hence $suv = p + u + v < 2(u + v) \leq 4 \max(u, v)$. Cancelling the maximum leaves a product of two of the variables below 4. But $\min(s, u, v)^3 > 2p \geq 8$ forces every variable to be at least 3, so that product is at least 9. $\square$

Theorem 9.3. $T(p, 5) = O_\varepsilon(p^{1/3+\varepsilon})$.

Proof. By Lemma 9.2 some variable is at most $(2p)^{1/3}$. If it is $s = t$, then $(tu - 1)(tv - 1) = tp + 1$ determines (u, v) from a divisor of $tp + 1$. If it is $u = t$ or $v = t$, then $v(su - 1) = p + u$ determines the pair from a divisor of $p + t$. Hence

$$C(p) \leq \sum_{1 \leq t \leq (2p)^{1/3}} (d(tp + 1) + 2d(p + t)),$$

and a divisor-sum bound of Shiu type [8] gives $\ll p^{1/3} \log p (\log \log p)^c$. The conclusion follows from $T(p, 5) = 5 + 5C(p)$ of Theorem 7.7. $\square$

For all 123 primes $5 \leq p < 700$ the ratio $T(p, 5)/\sqrt{p}$ has mean 15.2, and 12.0 over $p > 400$ (`code/t5_primes_split.py`); it does not decay on that range. Theorem 9.3 forces it to decay eventually, and the paragraph above places the onset near $p = 10^7$, so the small-prime data says nothing against it.

The conjecture is not new, and it is not ours. Conrey asked whether the number of positive integer solutions of $xyz + x + y = n$ is $O_\varepsilon(n^\varepsilon)$, and Ford generalised this to $xyz = A(x + y) + B$ with $O_\varepsilon(|AB|^\varepsilon)$ solutions; both questions are recorded in [9]. Our width-5 equation $guv = N(u + v) + M$ is Ford's equation with $A = N$ and $B = M$, so Conjecture 9.4 below is a case of Ford's problem, summed over $M \mid N^2$. Writing $R_a(n) = \#\{(x, y) \in \mathbb{N}^2 : axy - x - y = n\}$ for the quantity studied in [9], at a prime

$$C(p) = \sum_{a \geq 1} R_a(p), \quad R_1(p) = d(p + 1),$$

both verified for every prime $p < 200$. The second identity is Proposition 9.5: our lower bound is the $a = 1$ term. Huang proves an asymptotic for $\sum_{n \leq N} R_a(n)$ with a fixed, which is the averaged form of the inner sum.

That the frieze count is governed by a problem of Conrey and Ford is the reason Conjecture 9.4 is stated and not proved here.

Conjecture 9.4. $T(N, 5) = N^{o(1)}$.

The natural sharper guess, $T(N, 5) \asymp d(N^2) \log^2 N$, is false, and the reason is a lower bound that the numerics cannot see.

Proposition 9.5. $C(p) \geq d(p + 1)$, hence $T(p, 5) \geq 5d(p + 1)$, for every prime p . Consequently $T(N, 5) \asymp d(N^2) \log^2 N$ fails.

Proof. Taking $u = 1$ in (8), written as $suv = p + u + v$, leaves $sv = p + 1 + v$, which is solvable exactly when $v \mid p + 1$, with $s = (p + 1)/v + 1$. Distinct divisors give distinct solutions, so $C(p) \geq d(p + 1)$, and $T(p, 5) = 5 + 5C(p)$ by Theorem 7.7.

For the consequence, let m_r be the product of the first r primes, so $d(m_r) = 2^r$ and $\log m_r \sim r \log r$. By Linnik's theorem there is a prime $p \equiv -1 \pmod{m_r}$ with $\log p \ll \log m_r$. Then $m_r \mid p + 1$, so $d(p + 1) \geq 2^r$ while $\log^2 p \ll (r \log r)^2$, and $2^r/(r \log r)^2 \rightarrow \infty$. Since $d(p^2) = 3$ at a prime, the asserted equivalence would force $d(p + 1) = O(\log^2 p)$. $\square$

The crossover is far out of computational reach: 2^r overtakes $(r \log r)^2$ near $r = 20$, where m_r already exceeds 10^{25} . So the flat ratios observed for $N \leq 3000$ are what the proposition predicts, not evidence against it. This is why the equivalence are what the proposition predicts, not evidence against it: numerical stability over three decades is not an asymptotic.

Multiplying $auv = p + u + v$ through by a gives

$$(au - 1)(av - 1) = ap + 1, \tag{9}$$

so the solutions with leading coefficient a are the divisors of the shifted number $ap + 1$ lying in the class $-1 \pmod{a}$, and

$$C(p) = \sum_{a \geq 1} \#\{D \mid ap + 1 : D \equiv -1 \pmod{a}\}. \tag{10}$$

The term $a = 1$ is $d(p + 1)$, the same count as the $u = 1$ family of Proposition 9.5 though a different family. The term $a = 2$ is the one that costs nothing: $2p + 1$ is odd, so every divisor of it is odd and hence of the form $2u - 1$, and no divisor is lost to the congruence. Thus $R_2(p) = d(2p + 1)$ exactly, and since the two families carry different values of a they are disjoint.

Proposition 9.6. $C(p) \geq d(p + 1) + d(2p + 1)$.

Proof. Formula (9) at $a = 1$ and $a = 2$; the two families are disjoint because a is determined by the solution. Formalised as `cubic_lower_two_families`. $\square$

9.1 The count on the modular hyperbola

The decomposition (10) can be run in the other direction, eliminating a instead of u and v . Setting $d = au - 1$ and $e = av - 1$, identity (9) reads $de = ap + 1$, so $de \equiv 1 \pmod p$ and $a = (de - 1)/p$, while $a \mid d + 1$ and $a \mid e + 1$ are exactly the conditions defining u and v .

Proposition 9.7. $C(p) = \#\{(d, e) \in \mathbb{Z}_{>0}^2 : de \equiv 1 \pmod p, \frac{de-1}{p} \mid \gcd(d+1, e+1)\}$.

Proof. The maps $(a, u, v) \mapsto (au - 1, av - 1)$ and $(d, e) \mapsto ((de - 1)/p, (d + 1)p/(de - 1), (e + 1)p/(de - 1))$ are mutually inverse by (9). That the second is defined on the whole target set, which is where the divisibilities $(de - 1)/p \mid d + 1$ and $(de - 1)/p \mid e + 1$ are used, is checked rather than argued below, so this proposition rests on a computation at that point. Checked against $C(p)$ for all primes $5 \leq p < 120$ with no discrepancy (`code/modular_hyperbola.py`). $\square$

So $C(p)$ counts points of the modular hyperbola $de \equiv 1 \pmod p$ cut by a divisibility that ties the level $a = (de - 1)/p$ to $\gcd(d + 1, e + 1)$. The same substitution turns the sum that every upper bound passes through into a hyperbola count with no side condition,

$$\sum_{a \leq A} d(ap + 1) = \#\{(d, e) \in \mathbb{Z}_{>0}^2 : de \equiv 1 \pmod p, 1 < de \leq Ap + 1\}, \quad (11)$$

and this locates the wall. Counts of points of $de \equiv 1 \pmod p$ in regions are governed by Kloosterman sums, and Weil's bound gives them with an error $O_\varepsilon(p^{1/2+\varepsilon})$. With $A = (2p)^{1/3}$ the main term of (11) is of size $A \log p$, that is $p^{1/3+o(1)}$. The error term exceeds the main term. Kloosterman technology therefore does not reach the trivial bound $C(p) \ll_\varepsilon p^{1/3+\varepsilon}$ of Theorem 9.3, let alone improve on it, and the improvement would have to come from below the square root barrier rather than from a sharper application of Weil.

The substitution (9) is an identity over any commutative ring (`fixed_point_ring`), which is why the analogue over $\mathbb{F}_q[t]$ inherits the structure of the problem intact rather than simplifying it.

10 Formalization

The development accompanying this paper is a Lean 4 library against Mathlib. It contains no `sorry`, no `native_decide` and no additional axioms: every declaration cited below depends only on `propext`, `Classical.choice` and `Quot.sound`, which continuous integration checks on each commit.

What is machine-checked is the enumeration and the reductions it rests on. In particular the counts are taken over Definition 2.1 itself rather than over a convenient class: the definition is

transcribed with no periodicity assumed (periodicity is derived, not assumed, at widths 4 and 5), and the resulting sets are put in bijection with the arithmetic conditions the search enumerates. Theorem 4.1 and the divisibility statements of Section 6 are verified in that sense.

So is the count at a prime, end to end. Theorem 7.7 and Lemma 9.1 compose in the library into a single declaration giving $T(p, 5) = 5 + 5C(p)$ for every prime $p \geq 5$, stated over Definition 2.1 with periodicity derived rather than assumed, and with $C(p)$ the full solution count of $auv = p + u + v$ rather than a count inside a search box. Theorem 8.2 is verified as well, the support condition and the bijection at each M separately.

What is not machine-checked is stated as such. The external inputs of Shiu and of Linnik are cited, not proved; and the passages to O_ε in Theorem 9.3 and Corollary 7.3 are written proofs.

A per-statement table naming, for each result, the Lean declaration that verifies it or recording that none exists, is maintained with the library rather than here, since it changes with the library and not with the paper. Tables and computed constants in this paper are regenerated by the scripts named alongside them.

References

- [1] R. Zhang, *A positive Siegel theorem: Dynkin friezes and positive Mordell–Schinzel*, arXiv:2503.08800.
- [2] C. H. Conley and V. Ovsienko, *Quiddities of polygon dissections and the Conway–Coxeter frieze equation*, arXiv:2107.01234.
- [3] M. Cuntz and T. Holm, *Frieze patterns over integers and other subsets of the complex numbers*, J. Comb. Algebra **3** (2019), 153–188. arXiv:1711.03724.
- [4] J. H. Conway and H. S. M. Coxeter, *Triangulated polygons and frieze patterns*, Math. Gaz. **57** (1973), 87–94 and 175–183.
- [5] O. Karpenkov, I. Short, M. van Son and A. Zabolotskii, *Classifying integer tilings and hypertilings*, arXiv:2601.21445.
- [6] O. Karpenkov, *Geometry of Continued Fractions*, Algorithms and Computation in Mathematics **26**, Springer, 2013.
- [7] B. Conrey and N. Shah, *Which numbers are not the sum plus the product of three positive integers?*, arXiv:2112.15551.
- [8] P. Shiu, *A Brun–Titchmarsh theorem for multiplicative functions*, J. Reine Angew. Math. **313** (1980), 161–170.
- [9] J. Huang, *A mean value theorem for the Diophantine equation $axy - x - y = n$* , arXiv:1108.0095. The questions of Conrey and Ford quoted above are stated in its introduction.